Reach Before Price: When AI Testing Services Get Adopted, and Why Vendors Invest in Reach in Discrete Jumps

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Abstract

Vendors of AI-assisted software testing sell two things that buyers tend to run together: speed, meaning how fast a tool finds faults, and reach, meaning how large a share of the faults that a competent human team could eventually find the tool can find at all. We build a model that separates the two. A firm with heterogeneous stakes chooses a testing budget under a Goel–Okumoto fault-discovery process, and a monopolist sets a usage price and, in a second step, chooses how much to invest in reach. Adoption is a single interval of stakes. When the tool's reach falls short of the human baseline, the interval is bounded, and the highest-stakes firms stay with human testing under every pricing scheme in which the net transfer to the firm is bounded. When reach matches the baseline, the bound disappears. The vendor's profit is continuous in reach but has a kink at parity, and as a result the optimal investment jumps from none to parity at a cost threshold and then stays at parity over an interval of costs. A planner's threshold lies strictly above the vendor's, so a subsidy of limited size changes nothing for vendors in an intermediate cost range. All results are proved except where we say otherwise; the quantitative illustrations use uncalibrated parameters. Public data on coding agents and open-source defect reports are consistent with the model's building blocks but do not test it.

Keywords: AI-assisted testing; software reliability; nonhomogeneous Poisson process; technology adoption; vendor pricing; reach investment; economics of information systems

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Data and code availability

All code and derived data tables needed to reproduce the numerical results and figures are provided in the replication package. Raw public data (Public Jira Dataset, SWE-bench Verified, a public collection of

coding-agent trajectories, and a public replication package of an AI-coding-assistant adoption study) are available from their original sources; the package contains scripts and instructions to download them. No proprietary or firm-level data are used.

Use of generative AI tools

[The authors must describe here any use of generative AI tools in the preparation of the manuscript, code, or analysis, in line with the journal's policy, and confirm that they take responsibility for the content.]